Robotics: Sensors, Actuators and Control

Knowledge document 016 of 100

Introduction to Robotics

Robotics is the branch of engineering and science that deals with the design, construction, operation and use of robots. A robot is a programmable machine that can sense its environment, make decisions and carry out physical actions, either automatically or under human guidance. Robotics combines mechanical engineering, electronics, computer science and artificial intelligence. The word "robot" comes from the Czech word "robota", meaning forced labour, and was first used in the 1920 play R.U.R. (Rossum's Universal Robots) by the Czech writer Karel Capek. In 1942 the author Isaac Asimov described the famous Three Laws of Robotics in his science fiction.

History

The first industrial robot, Unimate, designed by George Devol and promoted by Joseph Engelberger, started work on a General Motors assembly line in New Jersey in 1961, lifting hot metal parts. Since then robots have become common in factories, hospitals, warehouses, farms and homes. The Mars rovers, the Roomba robot vacuum cleaner released in 2002, and humanoid robots such as Honda's ASIMO and Boston Dynamics' Atlas show the progress of the field.

The Three Parts of a Robot: Sense, Think, Act

Every robot has three essential functions. Sensing means gathering information with sensors. Thinking means processing that information in a controller, which may be a microcontroller such as Arduino or a computer running software and AI. Acting means producing motion or force with actuators. A power supply and a mechanical structure complete the robot.

Sensors Used in Robots

Sensors are devices that measure physical quantities and give robots awareness of themselves and their surroundings. Three common sensors used in robots are the ultrasonic distance sensor, which measures distance to obstacles using sound echoes; the camera, which gives vision for recognising objects and navigating; and the infrared (IR) sensor, which detects nearby objects or follows a line on the ground. Other important sensors include LiDAR, which uses laser pulses to build a 3D map of surroundings; the inertial measurement unit (IMU), which combines an accelerometer and gyroscope to measure motion and orientation; encoders, which count wheel or joint rotations; touch and force-torque sensors, which sense contact and pressure; GPS for outdoor position; temperature and gas sensors; and microphones for voice commands. Combining readings from several sensors is called sensor fusion.

What Is an Actuator?

An actuator is a component that converts energy, usually electrical, into physical movement. It is the muscle of the robot, producing the motion of wheels, legs, arms and grippers. Common types include electric motors, hydraulic actuators and pneumatic actuators. DC motors spin continuously and drive wheels. Servo motors rotate to a precise angle and are used in joints and grippers. Stepper motors move in exact small steps and are used in 3D printers and precise positioning. Hydraulic actuators use pressurised oil and provide very high force for heavy machines such as excavators. Pneumatic actuators use compressed air and are fast and simple, widely used for pick-and-place grippers. Newer designs include soft actuators and artificial muscles. The end of a robotic arm, which grips or works on objects, is called the end effector.

Control Systems

A control system decides how the actuators should move based on sensor feedback. In open-loop control, commands are sent without checking the result. In closed-loop (feedback) control, sensors continuously measure the output and the controller corrects errors. The most widely used controller is the PID controller, which stands for proportional, integral and derivative; it calculates a correction from the current error, the accumulated past error and the rate of change of the error. Robots also use path planning, kinematics (calculating joint angles to reach a position) and, increasingly, machine learning and computer vision. The Robot Operating System (ROS) is an open-source software framework widely used for robot development.

Types of Robots

Industrial robots such as robotic arms work in factories. Mobile robots move around, including wheeled robots and Automated Guided Vehicles in warehouses. Humanoid robots look like humans. Collaborative robots, called cobots, work safely beside people. Service robots include cleaning and delivery robots. Medical robots, like the da Vinci surgical system, assist surgeons. Space robots include planetary rovers and robotic arms on satellites and space stations.

What Is a Robotic Arm Used For in Factories?

A robotic arm is a mechanical arm, with joints similar to a human shoulder, elbow and wrist, controlled by a computer. In factories it performs repetitive, precise or dangerous tasks such as welding car bodies, spray painting, assembling electronic components, picking and placing items, packaging, palletising boxes, machine tending and quality inspection. Robotic arms work continuously with high accuracy and speed, improve product quality, reduce workplace injuries and handle heavy loads. The number of independent joints of an arm is called its degrees of freedom, and a typical industrial arm has six.

Robotics in India

India has a growing robotics sector. The automobile plants around Chennai, such as those in Sriperumbudur and Oragadam, use many welding and painting robots. Invento Robotics of Bengaluru built the humanoid receptionist robot Mitra, and Addverb Technologies, based in Noida, builds warehouse robots. DRDO developed Daksh, a remote-controlled robot for defusing bombs. ISRO's Chandrayaan-3 rover, Pragyan, was a robot exploring the Moon, and ISRO's Gaganyaan programme planned a humanoid robot named Vyommitra for an uncrewed test flight. Many Indian engineering colleges run robotics clubs and the annual national contest Robocon.

Challenges and Future

Challenges include high cost, safety around humans, handling unstructured environments, battery life and the impact on jobs. The future includes AI-driven robots that learn tasks from demonstration, humanoid robots for warehouses, surgical and agricultural robots, and swarm robotics.

Quick Facts

- An actuator converts energy into motion; examples are DC motors, servo motors and hydraulic or pneumatic cylinders.
- Common robot sensors include ultrasonic sensors, cameras, IR sensors, LiDAR and IMUs.
- Robotic arms in factories weld, paint, assemble, pick and place, and pack products.
- The first industrial robot, Unimate, began work at General Motors in 1961.
- The word robot comes from Karel Capek's 1920 play R.U.R.